Can we Control Systems with Less Data Transfer?

Applying Intermittent Control on Furuta Pendulum

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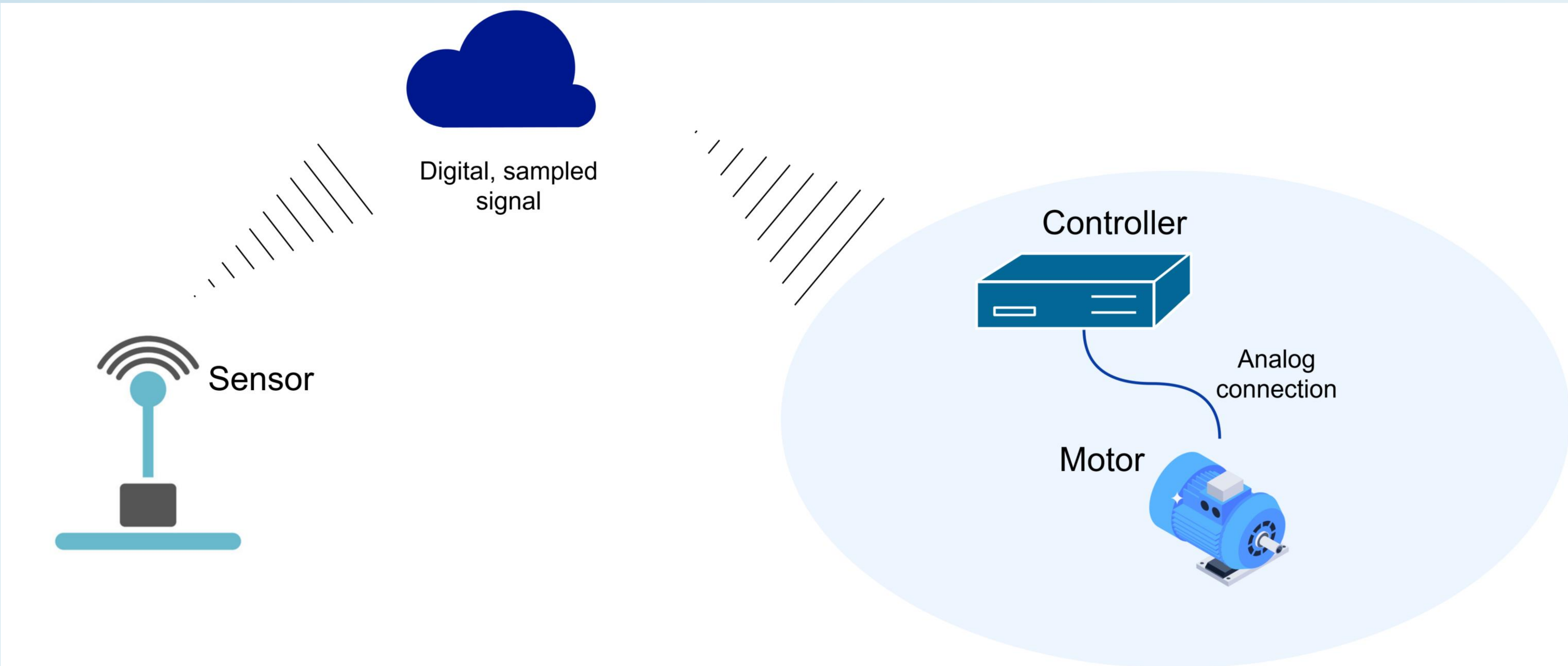
Yes!
By making the controller aware of the reference input and system dynamics, we can maintain performance **without continuous sensor sampling**, if no disturbances/noises occur.

We tested it on the underactuated **Furuta Pendulum**

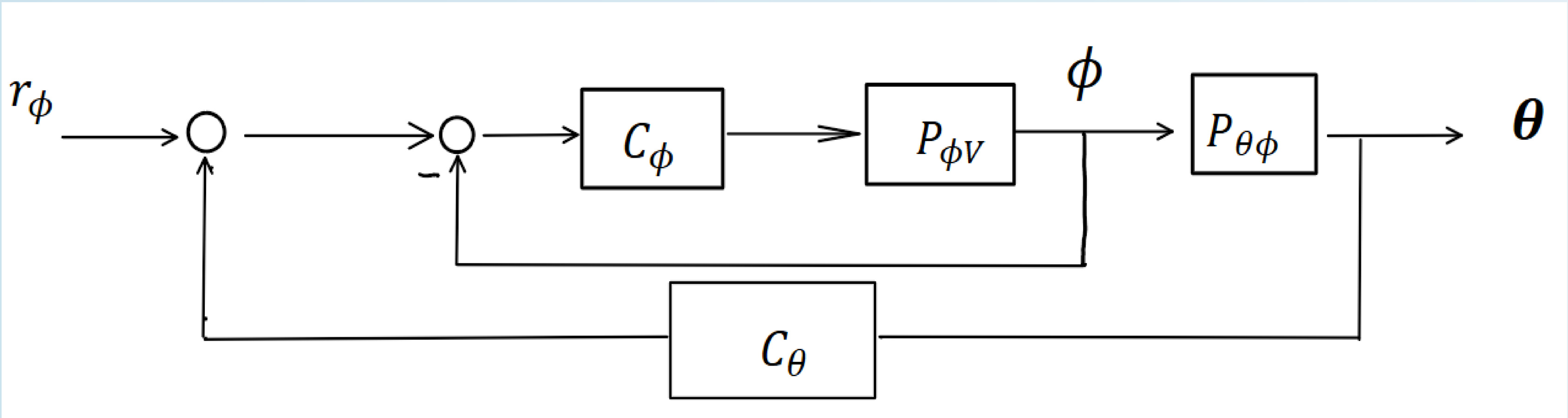


System & Tracking $\phi(t)$, in bottom eq.

The system has:
1. Input: voltage $V(t)$
2. Outputs: arm angle $\phi(t)$ and pendulum's angle $\theta(t)$
3. A linearized model, for bottom + top eq. is used

Tracking Loop: Control $\phi(t)$ using $C_\phi(s)$, $\phi(t)$ **oscillates when reaching the target**

Cascade Loop: Control $\phi(t)$ using $C_\phi(s)$ while damping $\theta(t)$ using $C_\theta(s)$
$\theta(t)$ **is damped, but at the cost of tracking** $r_\phi(t)$



Cascade Control block diagram

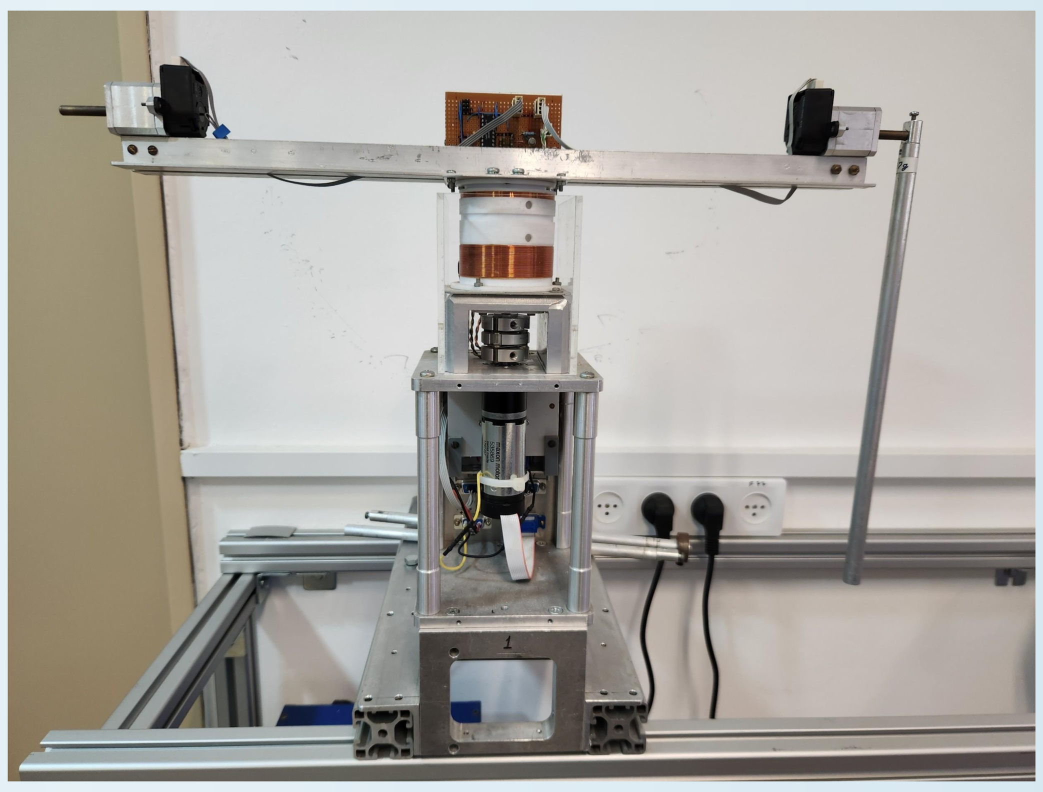
What is Intermittent control?

Purpose: Modify an existing controller with intermittent sensor communication.
Instead of standard ZOH, use a HOLD block that incorporate system dynamics + reference knowledge.

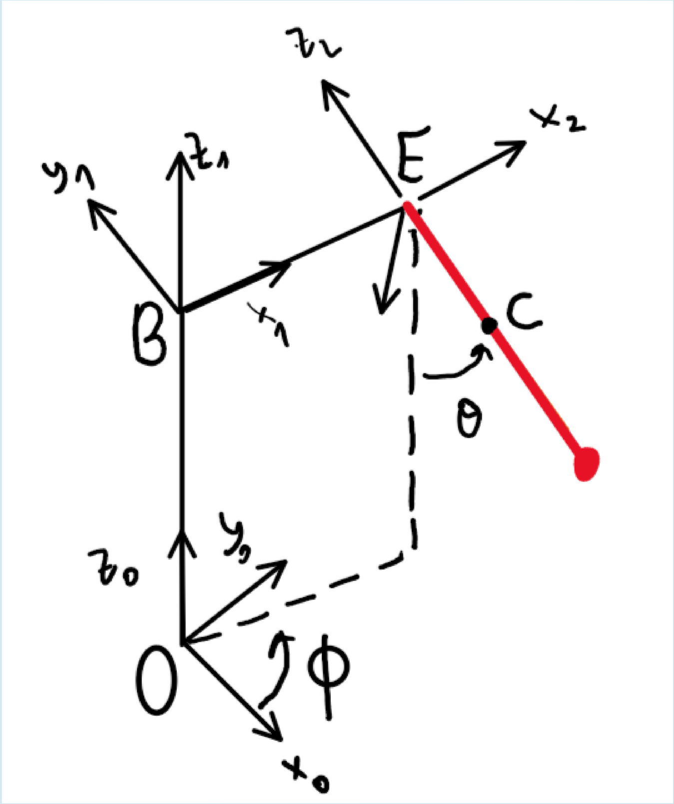
System Dynamics:

$$P \text{ with } r_\phi : \begin{cases} \dot{x}_w = A_w x_w + B_w u_1 + B_r r_\phi \\ \begin{pmatrix} \phi \\ \theta \end{pmatrix} = \begin{pmatrix} C_w \\ C_t \end{pmatrix} x_w \end{cases}$$

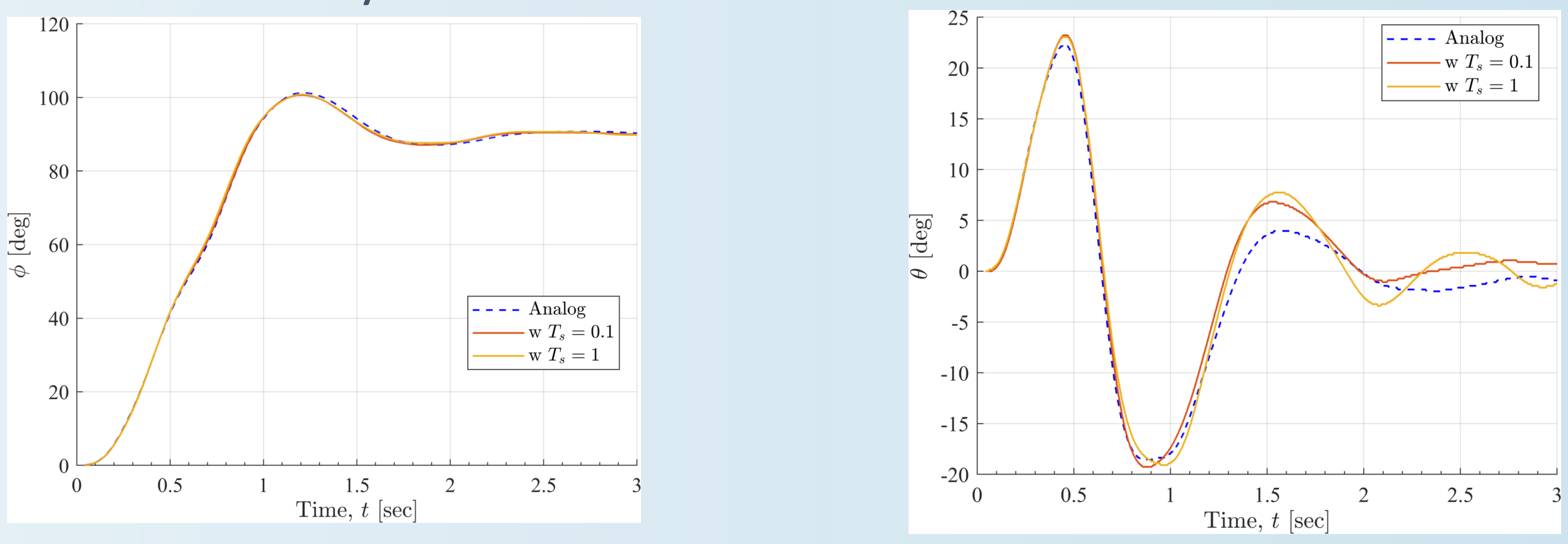
Example: A $T_s(t)$ changing with $\dot{\theta}(t)$ size has been used.



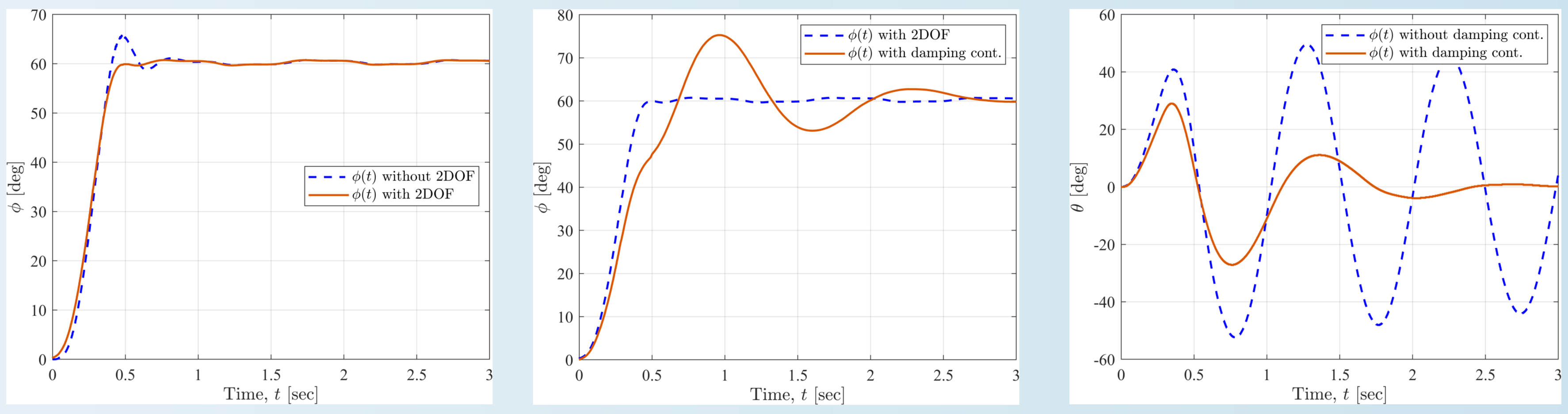
Furuta pendulum
the Lab system



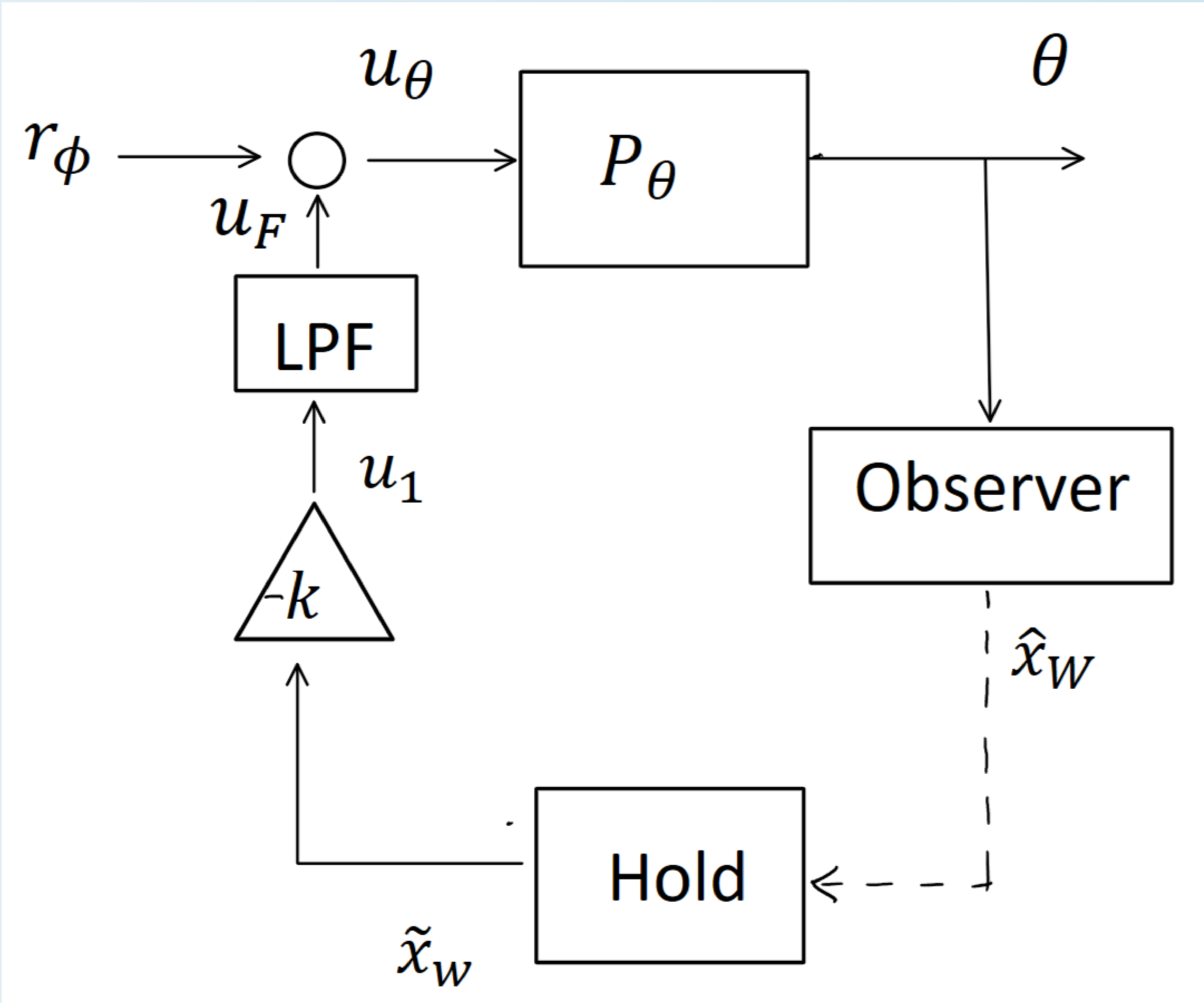
kinematic model



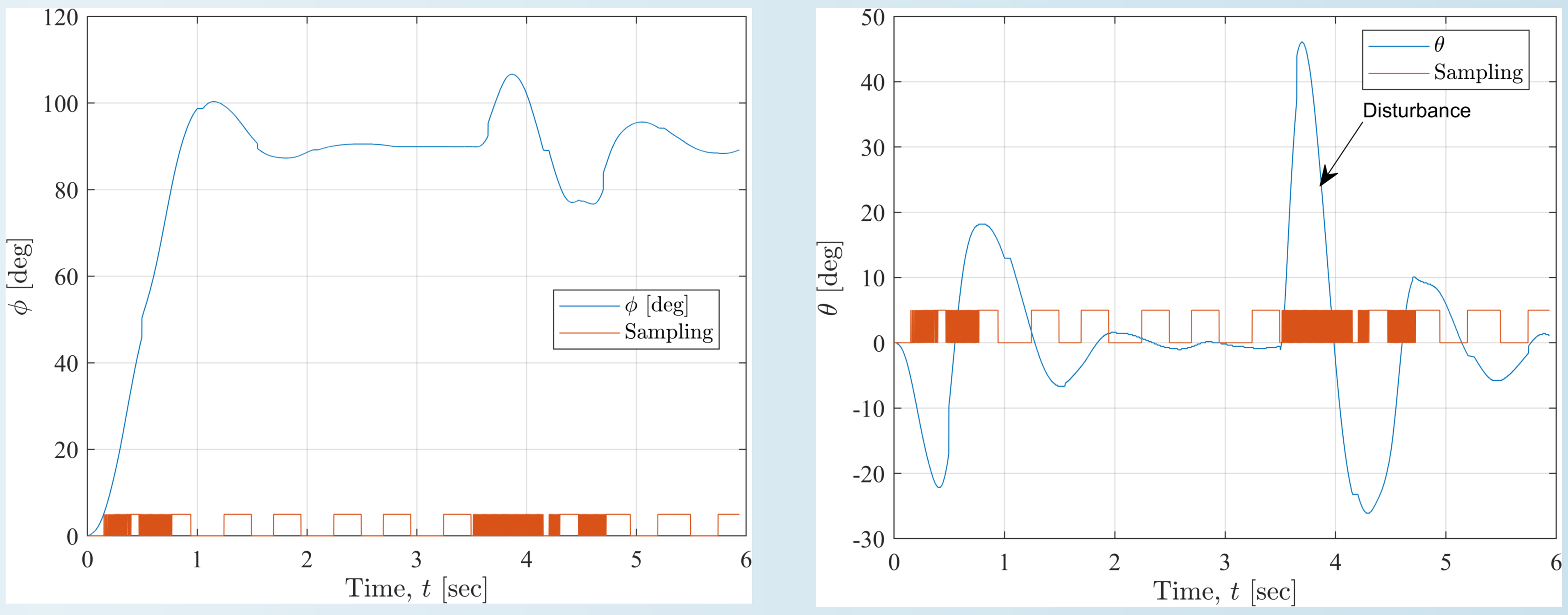
Tracking & Damping for int. control, static $T_s[s]$



Tracking and damping with cascade control



Controller in the
intermittent scheme



Tracking & Damping in int. control, with dynamic $T_s(t)[s]$